REVIEW OF THE OFFICIAL REVIEWER

on the dissertation work of Yesmukhamedov Nurmaganbet Seitkaliuly on the topic «Computer System for Processing and Analysis of Eye Data to Support Laser Coagulation in the Treatment of Diabetic Retinopathy», provided for the degree of Doctor of Philosophy (PhD) in the educational program: 8D06102 — Computer and Software Engineering

№	**Criteria**	**Eligibility (please select one answer option)**	**Justification of the official reviewer's position**
1	The topic of the thesis (as of the date of its approval) corresponds to the directions of scientific development and/or government programs	1.1 Compliance with priority areas of scientific development or government programs: 1) The dissertation was completed within the project or target program financed from the state budget (indicate the name and number of the project or program); 2) The dissertation was completed within the other state program (indicate the name of the program); <u>3) The thesis corresponds to the priority direction of scientific development, approved by the Higher Scientific and Technical Commission under the Government of the Republic</u>	The topic corresponds to the priority direction of scientific development «Information and communication technologies» approved by the Higher Scientific and Technical Commission under the Government of the Republic of Kazakhstan, and to the state priorities of healthcare digitalisation. The direction of the research corresponds to the following documents: 1. The Concept for the Development of Artificial Intelligence for 2024–2029. 2. The Address of the President of the Republic of Kazakhstan «Kazakhstan in the Era of Artificial Intelligence» of 8 September 2025. 3. The Law of the Republic of Kazakhstan «On Artificial Intelligence» No. 230-VIII of 17 November 2025. 4. Subparagraph 2 of paragraph 3 of Article 20 of the Law of the Republic of Kazakhstan «On Science», in accordance with which the work was carried out. The dissertation was not financed from the state budget and was not performed under a state programme or an institutional mandate, and the volume states this explicitly rather than leaving it to be inferred. I regard that precision as proper: the work claims a correspondence of direction and nothing more. The subject matter — automated

		<u>of Kazakhstan (indicate direction).</u>	screening for diabetic retinopathy under the conditions of rural and resource-limited healthcare — falls directly within the national eHealth agenda.
2	Importance for science	The work <u>makes</u> / does not make a significant contribution to science and its importance is <u>well-disclosed</u> / not disclosed.	The scientific importance of the work lies not in the application of convolutional networks to fundus photographs, which is by now an established practice, but in the object the candidate chose to put under experimental control. The dominant tradition treats image preprocessing as ancillary preparation of the data, on the assumption that a sufficiently deep network learns the required invariances from raw pixels; the candidate reframes preprocessing as a binding component of the diagnostic model, on the ground that it defines the feature space available to the network and therefore co-determines what the network can learn. What raises this above a programmatic statement is that the reframing is not asserted but placed under a controlled contrast against an equivalent configuration trained without the pipeline, on two architecture families, with the success criterion fixed before the experiment that tested it. A second element of importance is methodological in a different sense: the mechanism ordinarily *invoked* to explain gains in transfer — the reduction of the distance between the training distribution and the target distribution — is here *measured*, at the penultimate layer, under a stated protocol condition, and thereby converted from an explanatory habit into a falsifiable quantity. The importance of the work is disclosed in the volume clearly and without inflation.
3	The principle of self-reliance	Self-reliance level: <u>1) high</u>; 2) average; 3) low;	The formalisation of the eight-stage pipeline, the controlled experimental design, the statistical validation framework and the architecture of the screening system are the candidate's personal work, and the volume identifies the boundary of that work

		4) no independence.	explicitly, including the point at which one stage of the pipeline extends his own previously published results rather than originating in the dissertation. Two further indications weigh with me. The first is the acceptance gate for the initialisation of the integrated arm: the candidate did not assume a pretraining strategy but tested candidate initialisations with a frozen-backbone linear probe, and reports that label-free self-supervision trained from scratch on the in-domain corpus failed that gate — a negative result of his own, reported as evidence rather than omitted. The second is the analytic identification of a structural defect shared by three measures in common use for expressing external robustness, made in full awareness that the finding rehabilitates none of his own numbers. Both are marks of an independent researcher rather than of an executor. Of the five publications on the topic, two are led by the candidate as first author.
4	The principle of internal unity	4.1 Justification of the relevance of the thesis: <u>1) justified</u>; 2) partially justified; 3) not justified.	The relevance is justified from two sides at once, and neither is left as an assertion. Clinically: diabetic retinopathy is among the leading causes of preventable blindness in the working-age population, its early stages are asymptomatic, and screening coverage is limited by the number of ophthalmologists rather than by the availability of cameras — acutely so in rural and resource-limited settings. Methodologically: the volume identifies a specific gap, the absence of a formalised and experimentally controlled treatment of preprocessing as a component of the diagnostic model, and it is that gap, not the clinical motivation alone, that the work then addresses.
		4.2 The content of the thesis reflects its topic: <u>1) reflects</u>;	The topic names fundus image enhancement and CNN classification, and the content treats them as one object rather than as two consecutive steps: the eight-stage pipeline is specified as part

		2) partially reflects; 3) does not reflect.	of the model, and every experiment manipulates the integration rather than the preprocessing considered apart from the network it feeds. The process is followed in its entirety, from the image as it leaves the camera to the five-class grade the model assigns, which is precisely what the formulation of the topic promises.
		4.3 The purpose and objectives correspond to the topic of the thesis: <u>1) correspond</u>; 2) partially correspond; 3) do not correspond.	The aim — to develop and experimentally validate an integrated enhancement-and-classification framework and to establish what difference the specification makes under controlled contrast — is answered objective by objective: the problem domain (Chapter 1), the theoretical foundations (Chapter 2), the formalised methodology (Chapter 3), the experimental programme (Chapter 4), the reliability apparatus and limits (Chapter 5), the screening architecture (Chapter 6). Six objectives, six chapters, and no objective without a result reported against it.
		4.4 All sections and provisions of the dissertation are logically interconnected: <u>1) completely interconnected</u>; 2) the relationship is partial; 3) there is no relationship.	The volume is built as a single argument and reads as one. The structural feature I consider decisive is that every hypothesis is stated, with its acceptance criterion, *before* the experiment that tests it, and is answered by that experiment — including where the answer is partial, as with the explainability hypothesis, whose qualitative half was not carried out and is reported as not carried out. The negative and partial outcomes are carried through to the closing statements rather than dropped along the way, which is what makes the chain from problem to conclusion traceable in both directions.
		4.5 The new solutions (principles, methods) proposed by the author are argued and evaluated in comparison with known	The critical analysis of existing automated screening systems in Chapter 1 is the candidate's own comparative reading and not a compilation of quotations: the systems are compared, their limitations argued, and the end-to-end paradigm is criticised on a specific ground rather than in general terms. The work is

		solutions: 1) there is a critical analysis; 2) partial analysis; 3) the analysis does not represent one's own opinions, but quotes from other authors.	subsequently placed against published systems without being ranked among them, on the stated ground that no two of the compared results share their task definition, corpus, reference standard and metric at once. I regard that refusal to produce a ranking table as a considered methodological position and not as an omission.
5	The principle of scientific novelty	5.1 Are the scientific results and provisions new? 1) completely new; 2) partially new (25–75% are new); 3) not new (less than 25% are new).	The novelty is concentrated in the specification of the model rather than in the choice of network. I note in particular: the eight-stage pipeline formalised as a component of the diagnostic model, with the field-of-view mask carried into the network as a fourth input channel alongside RGB; the dual-constraint stochastic CLAHE on the LAB luminance channel, with the clip limit CL = min(clip_factor × tile_area / 256, global_threshold × tile_area) applied with 80 % probability at training time, so that one stage serves at once as an enhancement operator and as a source of regularisation; the adaptive flat-field correction whose Gaussian parameter scales with the per-image field-of-view diameter (σ = 0.07 × D) instead of a fixed global value, applied only inside the mask so that the padding is not corrupted; the explicit binary field-of-view mask introduced as a dedicated stage rather than as an implementation detail, which prevents padding artefacts from being learned as features; and the cumulative ablation performed under a single initialisation, which separates the contribution of preprocessing from the initialisation with which the factorial design necessarily confounds it. The volume states correctly that the CLAHE formulation extends the candidate's own published work and that what is new here is its formalisation, its embedding in a specified pipeline and its validation in the five-class grading task. Two further contributions

			are non-empirical by design and are declared as such: the coupled thermal-optical model of fundus tissue response under laser exposure is a theoretical and computational contribution, and the screening-system architecture is a design contribution.
		5.2 Are the conclusions of the dissertation new? <u>1) completely new</u>; 2) partially new (25–75% are new); 3) not new (less than 25% are new).	The conclusions are new in the same sense as the results that carry them, and two of them are new in a stronger sense. The first is that the contribution of the individual stages is separable and orderable at the resolution of groupings, with the two photometric stages leading — a statement about the internal structure of a preprocessing regime that the literature normally treats as a single undifferentiated step. The second is that the mechanism invoked to explain transfer behaviour can be measured directly rather than inferred from its consequence, and that when it is measured the reduction is present on every external corpus while its magnitude does not track any performance gain. Both conclusions follow from the work's own measurements and are stated at the strength those measurements support.
		5.3 Technical, technological, economic, or managerial decisions are: <u>1) new and justified</u>; 2) completely new; 3) partially new (25–75%); 4) not new (less than 25%).	The technical decisions are new and, more to the point, justified rather than adopted: the parameter values of the two photometric stages were established by sweeps with interior optima confirmed on held-out data, the initialisation of the integrated arm was settled by an acceptance gate instead of by assumption, and the input representation with four channels is argued from the artefact it prevents. The modular screening architecture is a new managerial and technological solution for the screening workflow, submitted as a design specification.
6	Validity of the main conclusions	All key findings <u>are based</u> / are not based on scientifically sound evidence or are	The conclusions rest on 5-fold patient-level stratified cross-validation with leakage control over approximately 35,126 graded EyePACS images, with every primary metric reported as mean ±

		reasonably well founded (for qualitative research and areas of training in the arts and humanities).	standard deviation across four measures — weighted F1, ROC-AUC, Cohen's Kappa with quadratic weights and accuracy — rather than on a single figure of merit. Statistical support is provided by McNemar and DeLong tests, bootstrap confidence intervals over at least 1000 resamples, Holm/Bonferroni correction for multiple comparisons and mixed-effects modelling across folds. In the principal comparison the cross-validation intervals of the two arms do not overlap on any of the four primary metrics. I attach separate weight to the fact that the success criteria were fixed before the experiments that tested them: a criterion written once the result is known can always be satisfied, and it is the ordering that makes the resulting classification informative.
7	Main provisions submitted for defense	The following questions must be answered for each provision separately: 7.1 Has the position been proven? 1) proven; 2) rather proven; 3) rather not proven; 4) not proven. 7.2 Is it trivial? 1) yes; 2) no. 7.3 Is it new? 1) yes; 2) no. 7.4 Application level: 1) narrow; 2) medium; 3) wide. 7.5 Is it proven in the article? 1) yes; 2) no.	Eleven provisions are submitted; each is answered below in the order 7.1 / 7.2 / 7.3 / 7.4 / 7.5. **Provision 1.** Preprocessing of fundus images is a formalisable and experimentally testable component of the diagnostic model rather than ancillary preparation of the data — the reframing from the end-to-end paradigm to the integrated paradigm. 7.1 rather proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. The provision is methodological and non-empirical, and the volume submits it at that strength: the results are consistent with it under the conditions tested and do not establish it universally.
			Provision 2. The integrated configuration dominates the baseline in five-class classification on EyePACS, on both backbones, under the conjunctive pre-registered criterion. 7.1 proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. The criterion was satisfied in all three of its components on both backbones: Δ weighted F1 +6.54 pp and +6.55 pp, Δ ROC-AUC

			+0.0320 and +0.0360, Δ Kappa +0.1129 and +0.1103; DeLong p = 0.0041 and 0.0028, McNemar p = 0.0057 and 0.0041, Holm-corrected 0.0082 and 0.0056; the arm × backbone interaction is not significant (p = 0.31), so the effect does not depend on the architecture. The provision is correctly stated as a property of the configuration as a whole, since the two arms differ in initialisation as well as in preprocessing.
			Provision 3. The dual-constraint CLAHE parameters and the flat-field σ each exhibit an interior optimum within the ranges examined, confirmed on held-out data. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. Both optima lie inside the ranges swept and were confirmed on held-out data (+0.0599 and +0.0574 respectively). The volume states that the optimal values are properties of this corpus and pipeline rather than portable constants, which is the correct reading and bounds the provision honestly.
			Provision 4. The contributions of the individual pipeline stages are separable, monotone across folds, and orderable with the two photometric stages leading. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. Each stage transition exceeded the between-fold dispersion of its level, monotonicity held in each of the five folds, and the cumulative gain 0.7538 → 0.8193 recovers the whole in-domain difference. The ordering is submitted at the resolution of groupings only, adjacent ranks lying within noise, and the volume says so.
			Provision 5. The integrated configuration reduces the distance between the training distribution and each external distribution in penultimate-layer feature space, on every external corpus examined. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes.

			The distance fell on all six external corpora with every interval excluding zero, and no target-corpus statistic enters the transform. The provision is submitted in direction only.
			Provision 6. A model trained on EyePACS with the pipeline transfers to APTOS 2019 without retraining, clearing the pre-registered generalisation ratio $G \geq 0.85$ and exceeding the baseline on every grade. 7.1 proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. $G = 0.8976$. The baseline also clears the threshold, so the criterion does not discriminate between the arms and the evidence lies in the comparison — a reading the volume makes itself.
			Provision 7. Model attention aligns more closely with expert pixel-level lesion annotation under the integrated configuration. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. Established on all four annotated lesion types and on both the primary and the secondary measure, robustly across the attention threshold. It is alignment between model evidence and expert annotation, not clinical localisation of pathology, on one annotated corpus and one fold.
			Provision 8. Classification performance is maintained across every camera grouping examined and the dispersion of performance across those groupings is reduced. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. All five groupings clear the retention floor for both arms, and the substantive result is the contraction of the between-group dispersion. Two of the five groupings are the external clinical corpora themselves and furnish no independent replication; none of this constitutes device certification.
			Provision 9. The integrated configuration attains higher absolute

			weighted F1 than the baseline on both external clinical corpora, each difference reaching the minimal clinically important difference with its interval excluding zero. 7.1 proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. 0.5938 → 0.6627 and 0.6282 → 0.6823, evaluated on each corpus separately. The provision is submitted as higher absolute external performance and expressly not as resistance to degradation: relative to their own in-domain levels the two configurations decline almost identically.
			Provision 10. Three measures in common use for expressing external robustness — the degradation quantity, the generalisation ratio and the retention ratio — share a structural defect: each references an arm's external performance to that same arm's in-domain performance and therefore penalises a configuration for its in-domain strength. 7.1 proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. The result is analytic and strictly descriptive, and it rehabilitates no result of this work.
			Provision 11. A coupled thermal-optical model of fundus tissue response under laser exposure, and a modular architecture for an automated screening system in resource-limited environments. 7.1 rather proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. Submitted as theoretical and design contributions respectively — the model is not clinically validated, and the architecture was neither prototyped nor field-tested. I note with approval that each provision is submitted with a qualification the volume treats as inseparable from it, and that clinical validity, readiness for unsupervised deployment, device certification and localisation of pathology are expressly excluded

			from the defence.
8	The principle of reliability. Reliability of sources and information provided	8.1 The choice of methodology justified or is it described in sufficient detail? 1) yes; 2) no.	The methodology is described to the level at which it can be repeated: five-class grading, Focal loss with $\gamma = 2$ and inverse-frequency weighting, input resolution 512×512, mixed precision enabled for one backbone and deliberately disabled for the other, patient-level stratification of the folds, and an ingestion protocol for external images. The choice of each element is argued rather than asserted, and the pipeline is specified stage by stage with its operators, parameters and order of application.
		8.2 The results of the thesis work were obtained using modern methods of scientific research and techniques for processing and interpreting data using computer technologies: 1) yes; 2) no.	The methods are contemporary and appropriate to the task: convolutional classification on two established backbone families, transfer and in-domain pretraining, gradient-based attention analysis against pixel-level annotation, a distributional measure over penultimate-layer features, and a statistical apparatus built on resampling. The computational envelope is stated rather than left implicit — a single 12 GB accelerator at batch size 16 — which is what makes the results reproducible outside a well-equipped laboratory.
		8.3 Theoretical conclusions, models, identified relationships and patterns are proven and confirmed by experimental research: 1) yes; 2) no.	The theoretical claims are carried into experiment rather than left standing. The cumulative ablation confirms the decomposition it asserts: every stage transition contributed more than the between-fold dispersion of its level, monotonicity held in each of the five folds, the two photometric stages lead (flat-field +0.0143, CLAHE +0.0125, together some 41 % of the total gain), and the ablation recovers the entire in-domain gain of +0.0655 under a single initialisation. Image-quality evidence is reported independently of the classifier through contrast-to-noise ratio, entropy and SSIM.
		8.4 Important statements are supported / partially	Each substantive statement in the volume is referenced, and the referencing is proportionate: recent sources where the field is

		supported / not supported by references to current and reliable scientific literature.	moving quickly, and older, settled ones for the image-processing operators. Where a claim originates with the candidate — as with the CLAHE formulation — the volume identifies his own prior publication as its source rather than presenting it as new in full.
		8.5 The literature sources used <u>are sufficient</u> / are not sufficient for the literature review.	The reference list comprises 107 sources and is sufficient for the review: current in the deep-learning parts and appropriately conservative in the clinical ones. I regard it as a mark of reliability rather than of weakness that the volume states three limits of its own apparatus: that correction for multiple comparisons is scoped to the experiment within which the comparisons were planned, so no dissertation-wide error rate is claimed; that several evaluations rest on the models of one fitted fold and their intervals therefore understate total uncertainty in a known direction; and that one experiment depends on a clinical corpus that cannot be redistributed and is therefore not externally reproducible.
9	Principle of practical value	9.1 The dissertation has theoretical significance: <u>1) yes</u>; 2) no.	The theoretical significance is the reframing itself — preprocessing as a component that co-determines the feature space available to the network — together with the formalisations it required: the specification of the eight-stage pipeline, the dual-constraint clip limit, the adaptive flat-field correction and the ALO explainability metric. A contribution of a different kind sits beside them: a mechanism hitherto invoked as an explanation is made measurable under a stated protocol condition, which renders the mechanistic claim falsifiable independently of the performance claims it explains.
		9.2 The thesis has practical significance and there is a high probability of applying	The immediate practical result is the pipeline itself: a fully specified, reproducible preprocessing regime defined stage by stage, with its implementation reproduced in an appendix and its

		the obtained results in practice: 1) yes; 2) no.	computational envelope stated — a single 12 GB accelerator at batch size 16 and 512 × 512 input, which is within reach of a regional diagnostic centre. This is what makes the result usable by others rather than merely reported. The modular screening architecture is applicable to rural and underserved regions of Kazakhstan; it must be read as a design specification — no prototype, no clinical deployment test, alignment by design on data protection, and the clinician retains the diagnosis.
		9.3 Are the suggestions for practice new? 1) completely new; 2) partially new (25–75%); 3) not new (less than 25%).	The proposals for practice are new as a whole: a preprocessing regime specified to the level of operators, parameters and order of application, together with an architecture that carries it into a screening workflow with store-and-forward telemedicine and physician-in-the-loop decision support. The parameter values remain properties of the corpora on which they were fixed and would require re-derivation for a new acquisition setting, which the volume says plainly.
10	Quality of writing and design	Quality of academic writing: 1) high; 2) average; 3) below average; 4) low.	The volume is set out on 264 pages and contains 42 tables, 26 figures and two diagrams, with 107 sources in the reference list. The notation is consistent throughout, the abbreviations are declared in a dedicated register, and the formatting conforms to the applicable state standard. The quality I would single out, however, is not typographic. The academic register of the text is disciplined: claims are made at the strength the evidence supports and carry their qualifications inline rather than in a distant section on limitations, negative and partial outcomes are carried through to the closing statements instead of being quietly dropped, and the boundary between what was established and what was designed is visible on every page where it matters. This is a difficult standard to keep across a volume of this

			size, and the candidate keeps it.
11	Comments and suggestions on the dissertation		*1. Stage 3 of the pipeline — the generation of the field-of-view mask and its introduction as a fourth input channel — is not isolated in the cumulative ablation: the corresponding level applies Stages 2 and 3 jointly, so the mask channel, which the author lists among the elements of scientific novelty, has no separate contribution estimate. The joint level is significant, but its significance is shared with the crop. It would be desirable to add a single-stage level, or, failing that, to state in the text of the chapter that no separate estimate exists for this element.* *2. The ablation is cumulative and the stage order is fixed, so the contribution of each stage is measured with all preceding stages already applied. The reported ranking is therefore conditional on the order and is not order-invariant. The volume is aware of this, but the reservation is carried mainly in the qualifications attached to the corresponding provision; it deserves an explicit sentence at the point where the ranking is first presented.* *3. Several of the evaluations — cross-dataset transfer, the explainability analysis and the external clinical evaluation — rest on the checkpoints of one fitted fold. The reported intervals therefore quantify the sampling of the evaluation corpus alone and not the variability of training. The author states this among the limitations, and states it correctly; I would nevertheless recommend that the fold-0 basis be named in the captions of the corresponding tables, so that the reader meets the reservation together with the number rather than several chapters later.* The comments listed above do not affect the substance of the results obtained and do not alter my overall positive assessment of the work.

12	The scientific level of the doctoral candidate's articles on the research topic (in the case of defending the dissertation in the form of a series of articles, the official reviewers will comment on the scientific level of each article)		The main results are published in five scientific works. 1. An article in a journal indexed by Scopus — *Eastern-European Journal of Enterprise Technologies*, 2025, Vol. 4, No. 9(136), pp. 79–88, quartile Q3, CiteScore 2025 = 2.5, 42nd percentile in the category Computer Science Applications (#590 of 1022) and 44th percentile in Control and Systems Engineering (#217 of 390). 2. A paper in Scopus-indexed international conference proceedings — *Procedia Computer Science*, 2025, Vol. 272, pp. 496–501 (the 3rd International Workshop on Digital Society, DS 2025, Istanbul, Türkiye). 3–5. Three articles in journals recommended by the Committee for Quality Assurance in Science and Higher Education of the Republic of Kazakhstan: *News of the National Academy of Sciences of the Republic of Kazakhstan, Physico-Mathematical Series*, 2025, Vol. 2(354), pp. 74–91; *Herald of the Kazakh-British Technical University*, 2025, No. 4(75), Vol. 22, pp. 119–130; *Herald of KazUTB*, 2024, Vol. 2, No. 27-740. The publications correspond to the topic of the dissertation and cover its principal directions — image enhancement, preprocessing and analysis of fundus images, the information-system architecture and the mathematical modelling of laser exposure on fundus tissues. Their scientific level meets the requirements established for the defence of a doctoral dissertation.
13	The decision of the official reviewer (according to paragraph 28 of the current Model Regulation)		To petition the Committee for the award of the degree of Doctor of Philosophy (PhD).

Conclusion

The presented dissertation for the degree of Doctor of Philosophy (PhD) by Yesmukhamedov Nurmaganbet Seitkaliuly on the topic «Computer System for Processing and Analysis of Eye Data to Support Laser Coagulation in the Treatment of Diabetic Retinopathy» is a completed independent piece of scientific research, which meets the requirements of the Rules for awarding degrees, and its author deserves to be petitioned before the Committee for the award of the degree of Doctor of Philosophy (PhD) in the educational program 8D06102 — Computer and Software Engineering.

Official reviewer:

Candidate of Technical Sciences, Associate Professor
Department of Computer Engineering
International Information Technology University, Almaty Bektemyssova Gulnara Umitkulovna